

हिन्दुस्तान पेट्रोलियम कॉर्पोरेशन लिमिटेड
Hindustan Petroleum Corporation Limited
(A Maharatna Company)

Participant ID	
Participant Name	
Test Center Name	
Test Date	16/08/2025
Test Time	2:00 PM - 4:30 PM
Subject	Chemical Engineer

Section : English Language

Q.1 After his retirement, he withdrew completely from public life. He refused interviews, declined to attend reunions, and avoided all social contact. His life was now spent in contemplation, far from the spotlight.
What is the best one-word substitute for such withdrawal from society?

- Ans ☒ A. Diplomacy
☒ B. Nomadism
☒ C. Seclusion
☒ D. Chauvinism

Question ID : 441009387257
Option 1 ID : 4410091514652
Option 2 ID : 4410091514651
Option 3 ID : 4410091514650
Option 4 ID : 4410091514653
Status : Answered
Chosen Option : C

Q.2 Select the proverb/phrase that can substitute the underlined word segment in the given sentence.

To get the best seats at the concert, we need to understand that one who arrives early will get the best seat.

- Ans ☒ A. in the middle of difficulty lies opportunity
☒ B. the early bird catches the worm
☒ C. in order to get where you want to go, you first have to leave where you are
☒ D. lead by example

Question ID : 441009459945
Option 1 ID : 4410091798421
Option 2 ID : 4410091798420
Option 3 ID : 4410091798422
Option 4 ID : 4410091798423
Status : Answered
Chosen Option : B

Q.3 Select the most appropriate option to fill in the blank. If there is no article required, select 'No article needed'.

The art collectors are searching for _____ unique art-piece.

Ans ☒ A. a

☒ B. No article needed

☒ C. an

☒ D. the

Question ID : 441009400054

Option 1 ID : 4410091565670

Option 2 ID : 4410091565673

Option 3 ID : 4410091565671

Option 4 ID : 4410091565672

Status : Answered

Chosen Option : C

Q.4 Select the INCORRECTLY spelt word in the given sentence.

Feel free to drizzle a teaspoon of honey or maple syrup over the top to make it more enticing.

Ans ☒ A. honey

☒ B. syrup

☒ C. enticing

☒ D. drizzle

Question ID : 441009460011

Option 1 ID : 4410091798677

Option 2 ID : 4410091798678

Option 3 ID : 4410091798679

Option 4 ID : 4410091798676

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.5 Select the ODD sentence out.

Ans ☒ A. The girls enjoyed playing in the park.

☒ B. The old house has been abandoned.

☒ C. Nancy is going to the store.

☒ D. Jane was singing a song.

Question ID : 441009400304

Option 1 ID : 4410091566666

Option 2 ID : 4410091566669

Option 3 ID : 4410091566668

Option 4 ID : 4410091566667

Status : Answered

Chosen Option : B

Q.6 Select the most appropriate **ANTONYM** of the highlighted word in the given sentence.

If we are too relaxed, we become lethargic, wasteful and languid.

- Ans ☒ A. Active
☒ B. Slothful
☒ C. Static
☒ D. Dormant

Question ID : 441009460222

Option 1 ID : 4410091799543

Option 2 ID : 4410091799542

Option 3 ID : 4410091799541

Option 4 ID : 4410091799540

Status : Answered

Chosen Option : A

Q.7 Select the most appropriate option to fill in the blank.

This book is _____ interesting than the last one I read.

- Ans ☒ A. little
☒ B. most
☒ C. more
☒ D. least

Question ID : 441009400130

Option 1 ID : 4410091565981

Option 2 ID : 4410091565979

Option 3 ID : 4410091565978

Option 4 ID : 4410091565980

Status : Answered

Chosen Option : C

Q.8 Select the correct spelling of the incorrectly spelt word in the given sentence.

Healthy dietary choices and regular exercise are among the most effective weapons in the fight against high cholasterol.

- Ans ☒ A. dietary/dietry
☒ B. weapon/ wepon
☒ C. healthy/helthy
☒ D. cholastral/cholesterol

Question ID : 441009460047

Option 1 ID : 4410091798817

Option 2 ID : 4410091798818

Option 3 ID : 4410091798816

Option 4 ID : 4410091798819

Status : Marked For Review

Chosen Option : A

Q.9 Select the most appropriate **ANTONYM** of the highlighted word in the given sentence.

The defendant is accused of attempting to conceal evidence.

- Ans ☒ A. Hide
☒ B. Cloak
☒ C. Reveal
☒ D. Wrap

Question ID : 441009460212

Option 1 ID : 4410091799493

Option 2 ID : 4410091799495

Option 3 ID : 4410091799492

Option 4 ID : 4410091799494

Status : Answered

Chosen Option : C

Q.10 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

- A. The MoD is responsible for coordinating and supervising all agencies and functions related to national security and the Armed Forces.
B. The Indian Armed Forces, comprising the Army, Navy, and Air Force, are the primary components of this system.
C. In India, this system is overseen by the Ministry of Defence, with the President as the Supreme Commander of the Armed Forces.
D. A national defence system encompasses the structures, policies, and resources a nation uses to protect its sovereignty, territorial integrity, and national interests from external threats.

- Ans ☒ A. ABCD
☒ B. DCBA
☒ C. BCDA
☒ D. DCAB

Question ID : 441009446258

Option 1 ID : 4410091743705

Option 2 ID : 4410091743706

Option 3 ID : 4410091743707

Option 4 ID : 4410091743708

Status : Answered

Chosen Option : D

Q.11 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

- A This steady flow carries with it oxygen, fuel, hormones, other compounds and a host of essential cells.
B. The heart beats about 2.5 billion times over the average lifetime, pushing millions of gallons of blood to every part of the body.
C. When the heart stops, essential functions fail, some almost instantly.
D. It also whisks away the waste products of metabolism.

- Ans ☒ A. DABC
☒ B. BADC
☒ C. ABCD
☒ D. CDBA

Question ID : 441009446302

Option 1 ID : 4410091743880

Option 2 ID : 4410091743878

Option 3 ID : 4410091743877

Option 4 ID : 4410091743879

Status : Answered

Chosen Option : B

Q.12 Select the option that can substitute the underlined word segment in the given sentence.

Starting a business during an economic crisis is like engaging in a risky or dangerous activity, often with unpredictable consequences.

- Ans ☒ A. putting off till doomsday
☒ B. putting the car before the horse
☒ C. pulling the devil by the tail
☒ D. putting a spoke in somebody's wheel

Question ID : 441009459904

Option 1 ID : 4410091798265

Option 2 ID : 4410091798267

Option 3 ID : 4410091798266

Option 4 ID : 4410091798268

Status : Answered

Chosen Option : C

Q.13 Even though she worked all night to prepare the report, she insisted her teammates take equal credit. She refused to take advantage of her leadership role or seek special recognition. Her behaviour created a culture of trust and collaboration in the team. What is the best one-word substitute for such a person?

- Ans ☒ A. Maverick
☒ B. Egalitarian
☒ C. Opportunist
☒ D. Bureaucrat

Question ID : 441009387262

Option 1 ID : 4410091514673

Option 2 ID : 4410091514670

Option 3 ID : 4410091514671

Option 4 ID : 4410091514672

Status : Not Answered

Chosen Option : --

MARGDARSHAN

Comprehension:

Read the given passage and answer the questions that follow.

Yesterday afternoon turned misty and cold. I was tempted to stay by the fire in my study instead of trudging through the muddy moor to Wuthering Heights. But after dinner, I saw the maid on her knees, surrounded by buckets and brushes, stirring up clouds of dust as she dumped cinders onto the fire. That sight was enough—I grabbed my hat and set off. After a four-mile walk, I reached the garden gate at Wuthering Heights just as the first light snowflakes began to fall.

At the top of the hill, the ground was frozen solid, and the air was so sharp it made me shiver. I couldn't open the chained gate, so I climbed over it and hurried up the stone path, lined with scraggly gooseberry bushes. I knocked again and again, until my knuckles hurt and the dogs began howling.

"Wretched people," I muttered to myself. "You deserve to be cut off from the rest of humanity, if you're this rude to visitors. I'd never leave my doors locked during the day. No matter what—they're letting me in." With that, I grabbed the latch and shook it hard.

At last, the sour-faced servant Joseph poked his head out of a nearby barn window and barked, "What do you want? The master's down at the stables. Go around the back if you need to speak to him."

SubQuestion No : 14

Q.14 Why did the narrator leave his house instead of staying by the fire?

- Ans ☒ A. The maid was cleaning and creating dust.
- ☐ B. The fire had gone out.
- ☐ C. He forgot something at Wuthering Heights.
- ☐ D. He had an appointment at Wuthering Heights.

Question ID : **441009383606**

Option 1 ID : **4410091500020**

Option 2 ID : **4410091500023**

Option 3 ID : **4410091500022**

Option 4 ID : **4410091500021**

Status : **Not Answered**

Chosen Option : --

Comprehension:

Read the given passage and answer the questions that follow.

Yesterday afternoon turned misty and cold. I was tempted to stay by the fire in my study instead of trudging through the muddy moor to Wuthering Heights. But after dinner, I saw the maid on her knees, surrounded by buckets and brushes, stirring up clouds of dust as she dumped cinders onto the fire. That sight was enough—I grabbed my hat and set off. After a four-mile walk, I reached the garden gate at Wuthering Heights just as the first light snowflakes began to fall.

At the top of the hill, the ground was frozen solid, and the air was so sharp it made me shiver. I couldn't open the chained gate, so I climbed over it and hurried up the stone path, lined with scraggly gooseberry bushes. I knocked again and again, until my knuckles hurt and the dogs began howling.

"Wretched people," I muttered to myself. "You deserve to be cut off from the rest of humanity, if you're this rude to visitors. I'd never leave my doors locked during the day. No matter what—they're letting me in." With that, I grabbed the latch and shook it hard.

At last, the sour-faced servant Joseph poked his head out of a nearby barn window and barked, "What do you want? The master's down at the stables. Go around the back if you need to speak to him."

SubQuestion No : 15

Q.15 What weather change happened just as the narrator arrived at Wuthering Heights?

- Ans ☐ A. The sun came out.
- ☐ B. A fog rolled in.
- ☒ C. Snowflakes started to fall.
- ☐ D. A rainstorm began.

Question ID : **441009383607**

Option 1 ID : **4410091500026**

Option 2 ID : **4410091500027**

Option 3 ID : **4410091500025**

Option 4 ID : **4410091500024**

Status : **Answered**

Chosen Option : **C**

Comprehension:

Read the given passage and answer the questions that follow.

Yesterday afternoon turned misty and cold. I was tempted to stay by the fire in my study instead of trudging through the muddy moor to Wuthering Heights. But after dinner, I saw the maid on her knees, surrounded by buckets and brushes, stirring up clouds of dust as she dumped cinders onto the fire. That sight was enough—I grabbed my hat and set off. After a four-mile walk, I reached the garden gate at Wuthering Heights just as the first light snowflakes began to fall.

At the top of the hill, the ground was frozen solid, and the air was so sharp it made me shiver. I couldn't open the chained gate, so I climbed over it and hurried up the stone path, lined with scraggly gooseberry bushes. I knocked again and again, until my knuckles hurt and the dogs began howling.

"Wretched people," I muttered to myself. "You deserve to be cut off from the rest of humanity, if you're this rude to visitors. I'd never leave my doors locked during the day. No matter what—they're letting me in." With that, I grabbed the latch and shook it hard.

At last, the sour-faced servant Joseph poked his head out of a nearby barn window and barked, "What do you want? The master's down at the stables. Go around the back if you need to speak to him."

SubQuestion No : 16

Q.16 What does the narrator's reaction to the locked gate and barking dogs reveal about his character?

Ans ☒ A. He is determined and not easily put off.

☐ B. He is impulsive and inconsiderate.

☐ C. He is used to being treated rudely.

☐ D. He is timid and easily discouraged.

Question ID : **441009383608**

Option 1 ID : **4410091500029**

Option 2 ID : **4410091500030**

Option 3 ID : **4410091500031**

Option 4 ID : **4410091500028**

Status : **Answered**

Chosen Option : **A**

Comprehension:

Read the given passage and answer the questions that follow.

Yesterday afternoon turned misty and cold. I was tempted to stay by the fire in my study instead of trudging through the muddy moor to Wuthering Heights. But after dinner, I saw the maid on her knees, surrounded by buckets and brushes, stirring up clouds of dust as she dumped cinders onto the fire. That sight was enough—I grabbed my hat and set off. After a four-mile walk, I reached the garden gate at Wuthering Heights just as the first light snowflakes began to fall.

At the top of the hill, the ground was frozen solid, and the air was so sharp it made me shiver. I couldn't open the chained gate, so I climbed over it and hurried up the stone path, lined with scraggly gooseberry bushes. I knocked again and again, until my knuckles hurt and the dogs began howling.

"Wretched people," I muttered to myself. "You deserve to be cut off from the rest of humanity, if you're this rude to visitors. I'd never leave my doors locked during the day. No matter what—they're letting me in." With that, I grabbed the latch and shook it hard.

At last, the sour-faced servant Joseph poked his head out of a nearby barn window and barked, "What do you want? The master's down at the stables. Go around the back if you need to speak to him."

SubQuestion No : 17

Q.17 Select the most appropriate ANTONYM of the given word as used in the passage

Scraggly

- Ans ☒ A. untidy
☒ B. neat
☒ C. thorny
☒ D. overgrown

Question ID : 441009383609

Option 1 ID : 4410091500034

Option 2 ID : 4410091500035

Option 3 ID : 4410091500033

Option 4 ID : 4410091500032

Status : Not Answered

Chosen Option : --

Section : Quantitative Aptitude

Q.1 Let x be a real number. The average value of seven x^2 's and three x 's is 19. Then find the value of 49 (the average value of squares of all possible values of x).

- Ans ☒ A. 1551
☒ B. 1250
☒ C. 1334.5
☒ D. 1422.5

Question ID : 441009402438

Option 1 ID : 4410091575143

Option 2 ID : 4410091575142

Option 3 ID : 4410091575144

Option 4 ID : 4410091575145

Status : Not Answered

Chosen Option : --

Q.2 A children play area is shaped like a rhombus and has area 31 sq m. If 60 m of fencing is needed to enclose the play area, the cost (in ₹), of laying electric wires along its two diagonals, at the rate of ₹250 per m, is:

- Ans ☒ A. 8,100
☒ B. 8,050
☒ C. 8,000
☒ D. 8,250

Question ID : 441009644051

Option 1 ID : 4410092530391

Option 2 ID : 4410092530392

Option 3 ID : 4410092530393

Option 4 ID : 4410092530390

Status : Not Answered

Chosen Option : --

Q.3 Basanti is 15 years younger than Sarayu. Eleven years later from now, Sarayu will be twice as old as Basanti. Find the present age of Basanti.

- Ans ☒ A. 5 years
☒ B. 3 years
☒ C. 4 years
☒ D. 2 years

Question ID : 441009563340

Option 1 ID : 4410092208480

Option 2 ID : 4410092208478

Option 3 ID : 4410092208479

Option 4 ID : 4410092208477

Status : Answered

Chosen Option : C

Q.4 A manufacturer offers a trade discount of 30% on a product whose marked price is ₹25,000. A dealer gets an additional cash discount of 10% on the net price. If the dealer sells the product for ₹18,000, find the profit percentage made by the dealer (rounded off to two decimal places).

- Ans ☒ A. 14.05%
☒ B. 15.05%
☒ C. 14.29%
☒ D. 15.25%

Question ID : 441009462443

Option 1 ID : 4410091808180

Option 2 ID : 4410091808181

Option 3 ID : 4410091808182

Option 4 ID : 4410091808183

Status : Not Answered

Chosen Option : --

Q.5 The radius of a circle is 14 cm. A sector of the circle with an angle 90° is folded to make a cone by joining its two radii. Find the total surface area of the cone.

$$\left\{ \text{Take } \pi = \frac{22}{7} \right\}$$

- Ans ☒ A. 214.5 sq.cm
☒ B. 201.6 sq.cm
☒ C. 158.2 sq.cm
☒ D. 192.5 sq.cm

Question ID : 441009462164

Option 1 ID : 4410091807080

Option 2 ID : 4410091807079

Option 3 ID : 4410091807077

Option 4 ID : 4410091807078

Status : Not Answered

Chosen Option : --

Q.6 A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank in the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 3 hours faster than the first pipe and 1 hour slower than the third pipe. The time required by the first pipe is:

- Ans ☒ A. 12 hours
☒ B. 15 hours
☒ C. 10 hours
☒ D. 6 hours

Question ID : 441009563257

Option 1 ID : 4410092208148

Option 2 ID : 4410092208145

Option 3 ID : 4410092208147

Option 4 ID : 4410092208146

Status : Not Answered

Chosen Option : --

Q.7 The total number of voters in a constituency is 6,40,000. In an election 72.5% of votes are casted. Winner, runner and others got 35.8%, 20.2% and 25% of casted votes respectively. 5% of casted votes are invalid. If the remaining are nota votes, then find the number of nota votes.

- Ans ☒ A. 81,200
☒ B. 56,800
☒ C. 64,960
☒ D. 74,560

Question ID : 441009459984

Option 1 ID : 4410091798573

Option 2 ID : 4410091798572

Option 3 ID : 4410091798574

Option 4 ID : 4410091798575

Status : Not Answered

Chosen Option : --

Q.8 What will be the simple interest on ₹5,000 for 2 years at an annual rate of 6%?

- Ans ☒ A. ₹550
☒ B. ₹500
☒ C. ₹700
☒ D. ₹600

Question ID : 441009362152
Option 1 ID : 4410091414825
Option 2 ID : 4410091414823
Option 3 ID : 4410091414826
Option 4 ID : 4410091414824
Status : Answered
Chosen Option : D

Q.9 A trader buys 70 kg of sugar at ₹35 per kg and mixes it with 50 kg of rice costing ₹40 per kg. He sells the mixture at ₹39 per kg. What is his overall profit in this transaction (in ₹)?

- Ans ☒ A. 210
☒ B. 220
☒ C. 250
☒ D. 230

Question ID : 441009462355
Option 1 ID : 4410091807841
Option 2 ID : 4410091807842
Option 3 ID : 4410091807844
Option 4 ID : 4410091807843
Status : Not Answered
Chosen Option : --

Q.10 The marks obtained by Rohit and Neha in an exam are in the ratio 4 : 7. If 4 marks are added to Rohit's score and 3 marks are subtracted from Neha's score, the new ratio becomes 3 : 5. Find the ratio of the sum of Neha's score and twice Rohit's score to the sum of Rohit's score and three times Neha's score.

- Ans ☒ A. 3 : 5
☒ B. 2 : 5
☒ C. 4 : 7
☒ D. 5 : 7

Question ID : 441009459931
Option 1 ID : 4410091798368
Option 2 ID : 4410091798369
Option 3 ID : 4410091798370
Option 4 ID : 4410091798371
Status : Not Answered
Chosen Option : --

Q.11 If $\tan\theta + \sec\theta = 24$, then find the sum of the digits in the value of $96\sec\theta$.

- Ans ☒ A. 15
☒ B. 10
☒ C. 14
☒ D. 11

Question ID : 441009644731

Option 1 ID : 4410092533041

Option 2 ID : 4410092533039

Option 3 ID : 4410092533042

Option 4 ID : 4410092533040

Status : Not Answered

Chosen Option : --

Q.12 Length, breadth and height of a cuboid are 10cm, 8cm and 3cm respectively. Let the total surface area of a cube be double the lateral surface of the given cuboid. If the side of the cube is doubled and measurements of cuboid are tripled, then find the sum of the lateral surface areas of new cube and cuboid.

- Ans ☒ A. 2105 cm²
☒ B. 1765 cm²
☒ C. 1645 cm²
☒ D. 1548 cm²

Question ID : 441009462472

Option 1 ID : 4410091808296

Option 2 ID : 4410091808297

Option 3 ID : 4410091808295

Option 4 ID : 4410091808298

Status : Not Answered

Chosen Option : --

Q.13 Mrunalini earned a profit of 25% by selling her diamond ring at a certain price. If she sells the ring at three times that original selling price, what would be her new profit percentage?

- Ans ☒ A. 275%
☒ B. 225%
☒ C. 220%
☒ D. 240%

Question ID : 441009462328

Option 1 ID : 4410091807720

Option 2 ID : 4410091807719

Option 3 ID : 4410091807717

Option 4 ID : 4410091807718

Status : Answered

Chosen Option : A

Q.14 The price of two different brands of adapters X and Y are in the ratio of 2: 3. X's price increased by 35% and the total price of X and Y together becomes ₹355, with an increase of 42%. By what percent the price of Y increased (rounded off to two decimal places)?

- Ans ☒ A. 47.75%
☒ B. 47.17%
☒ C. 46.67%
☒ D. 46.25%

Question ID : 441009462288
 Option 1 ID : 4410091807561
 Option 2 ID : 4410091807563
 Option 3 ID : 4410091807564
 Option 4 ID : 4410091807562
 Status : Not Answered
 Chosen Option : --

Q.15 The temperatures (in °C) recorded over seven days of the first week are 24.5, 32.5, 23.5, 28, 35, 3p, and q, while those for the second week are 26.5, 28.4, 20.6, 24, 27, 4p, and q. If the average temperatures of the first and second weeks are 29°C and 28°C respectively, find the values of p and q.

- Ans ☒ A. p=10, q=29.5
☒ B. p=19.5, q=29
☒ C. p=21, q=18.5
☒ D. p=20, q=28.5

Question ID : 441009402419
 Option 1 ID : 4410091575069
 Option 2 ID : 4410091575066
 Option 3 ID : 4410091575067
 Option 4 ID : 4410091575068
 Status : Not Answered
 Chosen Option : --

Q.16 If the equation $5x^2 - 3kx + 40 = kx^2 + 4$ has only one (repeated) solution then find the discriminant of $mx^2 - 2x - m = 0$.

- Ans ☒ A. 60
☒ B. 68
☒ C. -68
☒ D. -60

Question ID : 441009591556
 Option 1 ID : 4410092321342
 Option 2 ID : 4410092321343
 Option 3 ID : 4410092321344
 Option 4 ID : 4410092321341
 Status : Not Attempted and Marked For Review
 Chosen Option : --

Q.17 If $\frac{\left(\frac{-8}{3}\right)^{-8} \times \left(\frac{-3}{4}\right)^{10}}{\left(\frac{8}{3}\right)^9} = \frac{3^a}{2^b}$, then find the value of $b - a$.

- Ans ☒ A. 32
☒ B. 34
☒ C. 44
☒ D. 43

Question ID : 441009645079

Option 1 ID : 4410092534401

Option 2 ID : 4410092534398

Option 3 ID : 4410092534399

Option 4 ID : 4410092534400

Status : Not Answered

Chosen Option : --

Q.18 Let $2a + 3b = c$. Then find the value of $\frac{8a^3 + 27b^3 - c^3}{9abc} + \frac{4a^2 + 3b^2 - c^2}{b(4a + c)}$.

- Ans ☒ A. 8
☒ B. 5
☒ C. -4
☒ D. -6

Question ID : 441009462929

Option 1 ID : 4410091810290

Option 2 ID : 4410091810287

Option 3 ID : 4410091810289

Option 4 ID : 4410091810288

Status : Not Answered

Chosen Option : --

Q.19 Find the area of the sector when the radius of the circle is 78 cm, and the length of the arc is 6 cm.

- Ans ☒ A. 202 cm²
☒ B. 234 cm²
☒ C. 243 cm²
☒ D. 220 cm²

Question ID : 441009644680

Option 1 ID : 4410092532851

Option 2 ID : 4410092532854

Option 3 ID : 4410092532853

Option 4 ID : 4410092532852

Status : Not Answered

Chosen Option : --

Q.20 Kishore travelled a certain distance by train at the rate of 20 kmph. and walked back at the rate of 30 kmph. If the whole journey took 3 hours 20 minutes, then find the total distance travelled by him(in km).

- Ans ☒ A. 60
☒ B. 70
☒ C. 80
☒ D. 68

Question ID : 441009563305

Option 1 ID : 4410092208339

Option 2 ID : 4410092208338

Option 3 ID : 4410092208337

Option 4 ID : 4410092208340

Status : Answered

Chosen Option : C

Q.21 The LCM of two numbers is 300, and their HCF is 15. If one number is 60, what is the other number?

- Ans ☒ A. 75
☒ B. 45
☒ C. 120
☒ D. 90

Question ID : 441009985979

Option 1 ID : 4410093896290

Option 2 ID : 4410093896289

Option 3 ID : 4410093896292

Option 4 ID : 4410093896291

Status : Answered

Chosen Option : A

Q.22 The sum of three numbers is 381. If the ratio of the first to second is 5 : 9 and that of the second to the third is 2 : 11, then the positive difference of second and third numbers is:

- Ans ☒ A. 234
☒ B. 276
☒ C. 243
☒ D. 267

Question ID : 441009466051

Option 1 ID : 4410091822811

Option 2 ID : 4410091822812

Option 3 ID : 4410091822809

Option 4 ID : 4410091822810

Status : Not Answered

Chosen Option : --

Q.23 Two persons A and B started at a point 'O' and travelling in the directions of North-East and North-West with speeds $33\frac{1}{3}$ m/sec and 25m/sec respectively. Find the time when the distance between them is 300km.

- Ans ☒ A. 1.6 hours
☒ B. 2.5 hours
☒ C. 3.2 hours
☒ D. 2 hours

Question ID : 441009461963

Option 1 ID : 4410091806273

Option 2 ID : 4410091806276

Option 3 ID : 4410091806275

Option 4 ID : 4410091806274

Status : Not Answered

Chosen Option : --

Q.24 What is the value of x in the following equation?

$$19^{9.5} \times 19^{7.3} \times 19^{3.5} \div 19^{4.2} = 19^x$$

- Ans ☒ A. 18.1
☒ B. 16.1
☒ C. 15.1
☒ D. 17.1

Question ID : 441009358471

Option 1 ID : 4410091399885

Option 2 ID : 4410091399883

Option 3 ID : 4410091399882

Option 4 ID : 4410091399884

Status : Answered

Chosen Option : B

Q.25 A goods carrier covers a distance from place A to place B at the speed of 60 km/h and covers the distance from place B to place A at the speed of 90 km/h. What is the average speed of the goods carrier (in km/h)?

- Ans ☒ A. 75
☒ B. 74
☒ C. 71
☒ D. 72

Question ID : 441009563302

Option 1 ID : 4410092208328

Option 2 ID : 4410092208326

Option 3 ID : 4410092208325

Option 4 ID : 4410092208327

Status : Answered

Chosen Option : D

Q.26 Find the compound interest (in ₹) on ₹14,000 for 2 years at 5% per annum compounded annually.

- Ans ☒ A. 1,350
☒ B. 1,335
☒ C. 1,435
☒ D. 1,400

Question ID : 441009401969
Option 1 ID : 4410091573265
Option 2 ID : 4410091573262
Option 3 ID : 4410091573263
Option 4 ID : 4410091573264
Status : Answered
Chosen Option : C

Q.27 The cost of a dozen pens is ₹780 and that of 50 books is ₹2350. Then what is the cost of 3 dozens of books and 45 pens together?

- Ans ☒ A. ₹4617
☒ B. ₹3610
☒ C. ₹4050
☒ D. ₹5205

Question ID : 441009423009
Option 1 ID : 4410091653942
Option 2 ID : 4410091653943
Option 3 ID : 4410091653944
Option 4 ID : 4410091653945
Status : Answered
Chosen Option : D

Q.28 Pipes A and B can fill a tank in 4 and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then the tank will be filled in:

- Ans ☒ A. $3\frac{1}{5}$ hours
☒ B. $2\frac{3}{5}$ hours
☒ C. $2\frac{2}{5}$ hours
☒ D. $3\frac{2}{5}$ hours

Question ID : 441009563250
Option 1 ID : 4410092208118
Option 2 ID : 4410092208120
Option 3 ID : 4410092208119
Option 4 ID : 4410092208117
Status : Answered
Chosen Option : A

Q.29 Let x_1, x_2, x_3 and x_4 be the roots of fourth degree equation $2x^4 - 5x^3 - 149x^2 - 52x + 420 = 0$, then find the sum of the squares of the roots.

Ans ☒ A. 168.45

☒ B. 155.25

☒ C. 201.51

☒ D. 223.65

Question ID : 441009463035

Option 1 ID : 4410091810726

Option 2 ID : 4410091810724

Option 3 ID : 4410091810723

Option 4 ID : 4410091810725

Status : Not Answered

Chosen Option : --

Q.30 The arithmetic means of 48, 42, 52 and 58 observations are 26.5, 32.5, 40 and 35. Find the arithmetic mean of all 200 observations.

Ans ☒ A. 40.85

☒ B. 33.735

☒ C. 29.75

☒ D. 35.254

Question ID : 441009461428

Option 1 ID : 4410091804280

Option 2 ID : 4410091804278

Option 3 ID : 4410091804279

Option 4 ID : 4410091804277

Status : Not Answered

Chosen Option : --

Q.31 Sarayu and Hema are among 32 students who write an examination. Sarayu scores 101.5. The average score of the 31 students other than Hema is 72. The average score of all the 32 students is one more than the average score of the 31 students other than Sarayu. Find the score of Hema.

Ans ☒ A. 22

☒ B. 27

☒ C. 25

☒ D. 24

Question ID : 441009401642

Option 1 ID : 4410091571984

Option 2 ID : 4410091571987

Option 3 ID : 4410091571985

Option 4 ID : 4410091571986

Status : Not Answered

Chosen Option : --

Q.32 If after two successive discounts of 15% and 35%, an item is sold for ₹1105, what is its marked price (in ₹)?

- Ans ☒ A. 2,100
☒ B. 2,000
☒ C. 2,200
☒ D. 2,150

Question ID : 441009462413
Option 1 ID : 4410091808061
Option 2 ID : 4410091808062
Option 3 ID : 4410091808064
Option 4 ID : 4410091808063
Status : Answered
Chosen Option : B

Q.33 For a moderately skewed distribution, the mean of a dataset is 40 and the median is 45. If the mode of the distribution is k then find the mean of sum of the squares of first k natural numbers.

- Ans ☒ A. 1065
☒ B. 1036
☒ C. 1062
☒ D. 1063

Question ID : 441009466109
Option 1 ID : 4410091823042
Option 2 ID : 4410091823043
Option 3 ID : 4410091823044
Option 4 ID : 4410091823041
Status : Not Answered
Chosen Option : --

Q.34 The ratio of the length and the breadth of a rectangle is 6 : 7 and its area is 672 cm². Find the length of the rectangle (in cm).

- Ans ☒ A. 22
☒ B. 28
☒ C. 30
☒ D. 24

Question ID : 441009465959
Option 1 ID : 4410091822446
Option 2 ID : 4410091822445
Option 3 ID : 4410091822448
Option 4 ID : 4410091822447
Status : Not Answered
Chosen Option : --

Section : Intellectual Potential Test

Q.1 48 is related to 289 following a certain logic. Following the same logic, 32 is related to 193. To which of the following is 44 related to, following the same logic?
(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

Ans ☒ A. 267

☒ B. 265

☒ C. 273

☒ D. 271

Question ID : 441009390335

Option 1 ID : 4410091526914

Option 2 ID : 4410091526913

Option 3 ID : 4410091526916

Option 4 ID : 4410091526915

Status : Not Answered

Chosen Option : --

Q.2 Select the pair which follows the same pattern as that followed by the two set of pairs given below. Both pairs follow the same pattern.

QIV : TFY

MYM : PVP

Ans ☒ A. LYT : OVW

☒ B. TKB : WGE

☒ C. NMM : QJO

☒ D. PDP : SAR

Question ID : 441009363578

Option 1 ID : 4410091420566

Option 2 ID : 4410091420568

Option 3 ID : 4410091420569

Option 4 ID : 4410091420567

Status : Answered

Chosen Option : A

Q.3 Read the given statements and conclusions carefully. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which conclusion/s logically follow/s from the given statements
Statements: All discs are floppies. All floppies are pen drives. No pen drive is a hard drive.

Conclusions (I) : No disc is a hard drive

Conclusions (II) : Some pen drives are discs

Ans ☒ A. Neither conclusion (I) nor (II) follows

☒ B. Only conclusion (I) follows

☒ C. Both conclusions (I) and (II) follow

☒ D. Only conclusion (II) follows

Question ID : 441009363558

Option 1 ID : 4410091420488

Option 2 ID : 4410091420486

Option 3 ID : 4410091420489

Option 4 ID : 4410091420487

Status : Answered

Chosen Option : C

Q.4 If 'A' stands for '+', 'B' stands for 'x', 'C' stands for '+' and 'D' stands for '-', what will be come in place of question mark '?' in the following equation?
102 C 21 A 7 B 5 D 24 =?

Ans ☒ A. 93

☒ B. 92

☒ C. 94

☒ D. 91

Question ID : 441009388254

Option 1 ID : 4410091518608

Option 2 ID : 4410091518607

Option 3 ID : 4410091518609

Option 4 ID : 4410091518606

Status : Answered

Chosen Option : A

Q.5 If 2 is added to each odd digit and 1 is subtracted from each even digit in the number 73241685, what will be difference between the highest and lowest digits in the number thus formed?

Ans ☒ A. 5

☒ B. 7

☒ C. 8

☒ D. 6

Question ID : 441009388298

Option 1 ID : 4410091518774

Option 2 ID : 4410091518776

Option 3 ID : 4410091518777

Option 4 ID : 4410091518775

Status : Answered

Chosen Option : C

Q.6 Refer to the following letter, number, symbol series and answer the question
(Left) 8 & B A L N J N G E 9 \$ F ? Z 7 Y H 2 7 T Q (Right)
How many such numbers are there each of which is immediately preceded by a letter and also immediately followed by a symbol?

Ans ☒ A. Four

☒ B. Three

☒ C. One

☒ D. Two

Question ID : 441009360022

Option 1 ID : 4410091406081

Option 2 ID : 4410091406080

Option 3 ID : 4410091406078

Option 4 ID : 4410091406079

Status : Answered

Chosen Option : C

Q.7 Select the set in which the numbers are related in the same way as are the numbers of the following sets.
(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)
(32,158)
(38,188)

Ans ☒ A. (51,253)

☒ B. (41,202)

☒ C. (53,265)

☒ D. (63,318)

Question ID : 441009390300

Option 1 ID : 4410091526778

Option 2 ID : 4410091526777

Option 3 ID : 4410091526780

Option 4 ID : 4410091526779

Status : Marked For Review

Chosen Option : B

Q.8 If 'A' stands for '+', 'B' stands for 'x', 'C' stands for '+' and 'D' stands for '-', what will be come in place of question mark '?' in the following equation?
94 C 104 B 2 D 24 A 6 =?

Ans ☒ A. 296

☒ B. 290

☒ C. 298

☒ D. 292

Question ID : 441009388258

Option 1 ID : 4410091518624

Option 2 ID : 4410091518622

Option 3 ID : 4410091518625

Option 4 ID : 4410091518623

Status : Answered

Chosen Option : C

Q.9 Seven people H, I, J, K, L, C and D are sitting around a circular table facing the center of the table. J sits to the immediate right of K. Only two people sit between D and J when counted from the right of D. Only three people sit between I and K when counted from the right of K. H sits to the immediate right of L. Who sits fourth to the left of C?

Ans ☒ A. D

☒ B. K

☒ C. L

☒ D. I

Question ID : 441009372506

Option 1 ID : 4410091455651

Option 2 ID : 4410091455648

Option 3 ID : 4410091455649

Option 4 ID : 4410091455650

Status : Answered

Chosen Option : C

Q.10 Seven people P, Q, R, S, T, U and V are sitting around a circular table facing the centre. S is sitting 2nd to the right of U. T is an immediate neighbour of both P and Q. U is sitting 2nd to the right of P. R is not an immediate neighbour of P. The position of how many persons will remain unchanged if all persons are arranged in alphabetical order in clockwise direction starting from P?

- Ans ☒ A. Two
☒ B. One
☒ C. Three
☒ D. Four

Question ID : 441009466632
Option 1 ID : 4410091825292
Option 2 ID : 4410091825291
Option 3 ID : 4410091825293
Option 4 ID : 4410091825294
Status : Answered
Chosen Option : D

Q.11 In a certain code language, 'READ' is coded as '1346' and 'RUDE' is coded as '8316'. What is the code for 'A' in the given code language?

- Ans ☒ A. 1
☒ B. 3
☒ C. 6
☒ D. 4

Question ID : 441009466647
Option 1 ID : 4410091825353
Option 2 ID : 4410091825352
Option 3 ID : 4410091825354
Option 4 ID : 4410091825351
Status : Answered
Chosen Option : D

Q.12 What should come in place of ? in the given series based on the English alphabetical order?
OLP XUY GDH PMQ ?

- Ans ☒ A. VCD
☒ B. YVF
☒ C. YVZ
☒ D. VCF

Question ID : 441009360006
Option 1 ID : 4410091406024
Option 2 ID : 4410091406023
Option 3 ID : 4410091406022
Option 4 ID : 4410091406025
Status : Answered
Chosen Option : C

Q.13 Refer to the following letter, number, symbol series and answer the question

(Left) & & % J G 7 W ^ Y E @ 8 4 # 2 \ D 7 B N R L (Right)

How many such letters are there each of which is immediately preceded by a number and also immediately followed by a symbol?

Ans ☒ A. Three

☒ B. Two

☒ C. One

☒ D. Four

Question ID : 441009360020

Option 1 ID : 4410091406072

Option 2 ID : 4410091406071

Option 3 ID : 4410091406070

Option 4 ID : 4410091406073

Status : Answered

Chosen Option : C

Q.14 Each of F, G, H, I, M, N and O has an exam on a different day of a week starting from Monday and ending on Sunday of the same week.

Only two people have exam before M. Only one person has exam after F. Only three people have exam between M and I. Only one person has exam between O and H. G has exam immediately before O.

How many people have exam between N and O?

Ans ☒ A. One

☒ B. Four

☒ C. Three

☒ D. Two

Question ID : 441009359700

Option 1 ID : 4410091404802

Option 2 ID : 4410091404805

Option 3 ID : 4410091404804

Option 4 ID : 4410091404803

Status : Answered

Chosen Option : D

Q.15 Based on the English alphabetical order, three of the following four letter cluster pairs are alike in a certain way and thus form a group. Which is the one that does not belong to that group?

(Note: The odd man out is not based on the number of consonants/vowels or their position in the letter cluster)

Ans ☒ A. PM - QN

☒ B. LI - MJ

☒ C. RO - SK

☒ D. FC - GD

Question ID : 441009364596

Option 1 ID : 4410091424637

Option 2 ID : 4410091424635

Option 3 ID : 4410091424636

Option 4 ID : 4410091424634

Status : Answered

Chosen Option : C

Q.16 In a certain code language, 'SALE' is coded as '2168' and 'LACE' is coded as '7816'. What is the code for 'C' in the given code language?

- Ans ☒ A. 2
☒ B. 6
☒ C. 7
☒ D. 1

Question ID : 441009359725

Option 1 ID : 4410091404895

Option 2 ID : 4410091404897

Option 3 ID : 4410091404894

Option 4 ID : 4410091404896

Status : Answered

Chosen Option : C

Q.17 In the following triad, each group of letters is related to the subsequent one following a certain logic. Select from the given options, the one which follows the same logic.

MOVE – EVOM – EOVM
EARN – NRAE – NARE

- Ans ☒ A. IRON – RION – NOIR
☒ B. MAKE – EKAM – EAKM
☒ C. PAIN – PNAI – IAPN
☒ D. NICE – ICEN – NECI

Question ID : 441009466776

Option 1 ID : 4410091825872

Option 2 ID : 4410091825871

Option 3 ID : 4410091825870

Option 4 ID : 4410091825869

Status : Answered

Chosen Option : B

Q.18 In the following triad, each group of letters is related to the subsequent one following a certain logic. Select from the given options, the one which follows the same logic.

RING - GNRI - RNIG
BIND - DNBI - BNID

- Ans ☒ A. HAND - DNHA - HNAD
☒ B. BOND - ONBD - ONDB
☒ C. FARM - FRAM - MRAF
☒ D. REST - TSRE - ESRT

Question ID : 441009359746

Option 1 ID : 4410091404978

Option 2 ID : 4410091404981

Option 3 ID : 4410091404979

Option 4 ID : 4410091404980

Status : Answered

Chosen Option : A

Q.19 What should come in place of '?' in the given series?
3 4 12 39 103 ?

- Ans ☒ A. 228
☒ B. 230
☒ C. 224
☒ D. 226

Question ID : 441009390190
Option 1 ID : 4410091526335
Option 2 ID : 4410091526336
Option 3 ID : 4410091526333
Option 4 ID : 4410091526334
Status : Answered
Chosen Option : A

Q.20 Based on the English alphabetical order, three of the following four letter cluster pairs are alike in a certain way and thus form a group. Which is the one that does not belong to that group?
(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster)

- Ans ☒ A. KK-EI
☒ B. YY-WS
☒ C. ZZ – XT
☒ D. XX-VR

Question ID : 441009359989
Option 1 ID : 4410091405955
Option 2 ID : 4410091405954
Option 3 ID : 4410091405957
Option 4 ID : 4410091405956
Status : Answered
Chosen Option : A

Q.21 In a certain code language,
A + B means 'A is the son of B'
A – B means 'A is the wife of B'
A x B means 'A is the brother of B'
A ÷ B means 'A is the mother of B'
Based on the above, how is A related to K if 'A+T-HxS÷K'?

- Ans ☒ A. Mother's brother's son
☒ B. Father's brother
☒ C. Mother's brother
☒ D. Father's brother's son

Question ID : 441009466663
Option 1 ID : 4410091825417
Option 2 ID : 4410091825416
Option 3 ID : 4410091825415
Option 4 ID : 4410091825418
Status : Answered
Chosen Option : D

Q.22 Based on the alphabetical order, three of the following four are alike in a certain way and thus form a group. Which is the one that does not belong to that group?

(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster)

- Ans ☒ A. QNJ
☒ B. NKG
☒ C. JGC
☒ D. TQN

Question ID : 441009466765

Option 1 ID : 4410091825823

Option 2 ID : 4410091825825

Option 3 ID : 4410091825824

Option 4 ID : 4410091825826

Status : Answered

Chosen Option : D

Q.23 A, B, C, M, N and O live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it, number 2 and so on till the topmost floor is numbered 6. B lives on an even numbered floor but not on floor number 4. Only two people live between B and A. N lives on an odd numbered floor but not on the lowermost floor. Only two people live between N and C. M lives immediately below N. How many people live between O and M?

- Ans ☒ A. One
☒ B. Three
☒ C. Two
☒ D. Four

Question ID : 441009360571

Option 1 ID : 4410091408298

Option 2 ID : 4410091408300

Option 3 ID : 4410091408299

Option 4 ID : 4410091408301

Status : Answered

Chosen Option : C

Q.24 Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers

into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying etc. to 13

can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1

and 3 is not allowed)

(25,151)

(23,139)

- Ans ☒ A. (33,196)
☒ B. (31,185)
☒ C. (27,161)
☒ D. (28,169)

Question ID : 441009390306

Option 1 ID : 4410091526802

Option 2 ID : 4410091526801

Option 3 ID : 4410091526804

Option 4 ID : 4410091526803

Status : Answered

Chosen Option : D

Q.25 What should come in place of ? in the given series based on the English alphabetical order?
TVM PQP LLS ? DBY

- Ans ☒ A. GHK
☒ B. LHJ
☒ C. JHK
☒ D. HGV

Question ID : 441009364639
Option 1 ID : 4410091424809
Option 2 ID : 4410091424808
Option 3 ID : 4410091424807
Option 4 ID : 4410091424806
Status : Answered
Chosen Option : D

Q.26 Refer to the following letter, symbol series and answer the question. Counting to be done from left to right only.
(Left) K @ L & M H Q I £ \$ P & O * D E % B J G # F (Right)
If all the letters are dropped from the series, which of the following will be second from the left?

- Ans ☒ A. \$
☒ B. &
☒ C. £
☒ D. %

Question ID : 441009486081
Option 1 ID : 4410091899693
Option 2 ID : 4410091899695
Option 3 ID : 4410091899696
Option 4 ID : 4410091899694
Status : Answered
Chosen Option : B

Q.27 What should come in place of '?' in the given series?
3 4 9 28 ? 566

- Ans ☒ A. 114
☒ B. 111
☒ C. 113
☒ D. 112

Question ID : 441009390220
Option 1 ID : 4410091526464
Option 2 ID : 4410091526461
Option 3 ID : 4410091526463
Option 4 ID : 4410091526462
Status : Not Answered
Chosen Option : --

Q.28 Six people I, J, K, L, C and D live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it, number 2 and so on till the topmost floor is numbered 6. I lives on an odd numbered floor but not on the lowermost floor. J lives on an even numbered floor but not on floor number 4. Only two people live between J and C. Only two people live between I and L. D lives immediately below I. How many people live between K and D?

- Ans ☒ A. Three
☒ B. Two
☒ C. Four
☒ D. One

Question ID : 441009372546

Option 1 ID : 4410091455801

Option 2 ID : 4410091455802

Option 3 ID : 4410091455800

Option 4 ID : 4410091455803

Status : Answered

Chosen Option : B

Q.29 Refer to the following letter, symbol series and answer the question. Counting to be done from left to right only.
 (Left) Z C W J A @ Y W V O % % T M % @ * F = U Q (Right)
 How many such letters are there each of which is immediately preceded by a symbol and also immediately followed by a letter?

- Ans ☒ A. Five
☒ B. Four
☒ C. Two
☒ D. Three

Question ID : 441009364696

Option 1 ID : 4410091425034

Option 2 ID : 4410091425035

Option 3 ID : 4410091425037

Option 4 ID : 4410091425036

Status : Answered

Chosen Option : D

Q.30 In a certain code language,
 A+ B means 'A is the daughter of B'
 A - B means 'A is the brother of B'
 A x B means 'A is the wife of B'
 A # B means 'A is the father of B'
 Based on the above, how is H related to I if 'HxE-R+O#I'?

- Ans ☒ A. Brother's wife
☒ B. Wife
☒ C. Sister
☒ D. Brother's daughter

Question ID : 441009359718

Option 1 ID : 4410091404866

Option 2 ID : 4410091404868

Option 3 ID : 4410091404869

Option 4 ID : 4410091404867

Status : Answered

Chosen Option : A

Q.31 If 'A' stands for '+', 'B' stands for 'x', 'C' stands for '+' and 'D' stands for '-', what will be come in place of question mark '?' in the following equation?
81 A 9 C 86 D 12 B 4 =?

Ans ☒ A. 47

☐ B. 50

☐ C. 55

☐ D. 52

Question ID : 441009388259

Option 1 ID : 4410091518626

Option 2 ID : 4410091518627

Option 3 ID : 4410091518629

Option 4 ID : 4410091518628

Status : Answered

Chosen Option : A

Q.32 Mr. Opq starts from Point A and drives 9 km towards South. He then takes a right turn, drives 16 km, turns right and drives 25 km. He then takes a right turn and drives 5 km, turns right drives 9 km. He then takes a left turn drives 16 km. He takes a final right turn, drives 7 km and stops at Point P. How far (shortest distance) and towards which direction should he drive in order to reach Point A again? (All turns are 90 degrees turns only unless specified)

Ans ☐ A. 9 km to the north

☐ B. 7 km to the east

☒ C. 5 km to the west

☐ D. 8 km to the south

Question ID : 441009364666

Option 1 ID : 4410091424917

Option 2 ID : 4410091424915

Option 3 ID : 4410091424914

Option 4 ID : 4410091424916

Status : Not Answered

Chosen Option : --

Q.33 In this question, three statements are given, followed by two conclusions numbered I and II. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusion(s) logically follows/follow from the statements.

Statements:

All gases are liquids.

Some liquids are solids.

All solids are trees.

Conclusions:

I. All gases are solids.

II. Some trees are liquids.

Ans ☒ A. Only conclusion II follows.

☐ B. Both conclusions I and II follow.

☐ C. Neither conclusions I nor II follows.

☐ D. Only conclusion I follows

Question ID : 441009466676

Option 1 ID : 4410091825468

Option 2 ID : 4410091825469

Option 3 ID : 4410091825470

Option 4 ID : 4410091825467

Status : Answered

Chosen Option : D

Q.34 Seven people H, I, J, K, L, C and D are sitting in a row, facing north. C sits to the immediate left of K. I sits to the immediate right of J. Only two people sit to the right of K. Only two people sit between K and L. Only two people sit between H and C. How many people sits between D and I?

- Ans ☒ A. Four
☒ B. One
☒ C. Two
☒ D. Three

Question ID : 441009372518
Option 1 ID : 4410091455692
Option 2 ID : 4410091455695
Option 3 ID : 4410091455694
Option 4 ID : 4410091455693
Status : Answered
Chosen Option : B

Section : Domain Knowledge

Q.1 To geologically store CO₂, it must first be compressed, usually to a dense fluid state known as _____.

- Ans ☒ A. plasma
☒ B. anticline
☒ C. supercritical
☒ D. cleats

Question ID : 441009562853
Option 1 ID : 4410092206457
Option 2 ID : 4410092206454
Option 3 ID : 4410092206456
Option 4 ID : 4410092206455
Status : Not Answered
Chosen Option : --

Q.2 For a real gas, the entropy change between two states (1 and 2) at the same composition can be expressed as: $\Delta S = \Delta S_{ig} + \Delta S_R$, where ΔS_{ig} and ΔS_R represent ideal gas and residual contributions. Which of the following best explains the physical significance of ΔS_R ?

- Ans ☒ A. It accounts for the intermolecular forces and volume excluded effects in the real gas.
☒ B. It is negligible for gases at all pressures and temperatures.
☒ C. It measures the contribution of molecular vibrations and rotations to entropy.
☒ D. It corrects for the deviation from the ideal gas caused by molecular mass differences.

Question ID : 441009459721
Option 1 ID : 4410091797554
Option 2 ID : 4410091797556
Option 3 ID : 4410091797555
Option 4 ID : 4410091797553
Status : Answered
Chosen Option : A

Q.3 When working with Aspen Plus, what does a 'stream' typically represent in the simulation environment?

- Ans ☒ A. A calculation block
- ☒ B. A thermodynamic property
- ☒ C. A flow of material or energy
- ☒ D. A piece of equipment

Question ID : 441009644926

Option 1 ID : 4410092533790

Option 2 ID : 4410092533789

Option 3 ID : 4410092533788

Option 4 ID : 4410092533787

Status : Answered

Chosen Option : C

Q.4 Why is the batch reactor an example of an unsteady-state process?

- Ans ☒ A. Concentration of the components in the reactor would change with time
- ☒ B. Volume of reaction mixture decreases with extent of reaction
- ☒ C. Inlet flow rate of components in the reactor would change with time
- ☒ D. Non-ideal behaviour increases with time

Question ID : 441009512999

Option 1 ID : 4410092007039

Option 2 ID : 4410092007037

Option 3 ID : 4410092007036

Option 4 ID : 4410092007038

Status : Answered

Chosen Option : A

Q.5 For a two-stage adiabatic compressor, what variable is typically optimized to minimise the total theoretical work required?

- Ans ☒ A. Intermediate stage pressure.
- ☒ B. Inlet pressure of the gas.
- ☒ C. Inlet gas temperature.
- ☒ D. Outlet pressure from the second stage.

Question ID : 441009626208

Option 1 ID : 4410092458986

Option 2 ID : 4410092458984

Option 3 ID : 4410092458987

Option 4 ID : 4410092458985

Status : Answered

Chosen Option : C

Q.6 What does 'carbon neutrality' mean?

- Ans ☒ A. Using only carbon-based fuels
- ☒ B. Balancing emitted and offset carbon emissions
- ☒ C. Replacing carbon with nitrogen
- ☒ D. Emitting no carbon at all

Question ID : 441009619449

Option 1 ID : 4410092431947

Option 2 ID : 4410092431949

Option 3 ID : 4410092431950

Option 4 ID : 4410092431948

Status : Answered

Chosen Option : D

Q.7 What kind of dataset is best suited for unsupervised learning?

- Ans ☒ A. Data that is balanced across all classes
- ☒ B. Data without labelled outputs
- ☒ C. Data with clearly labelled output classes
- ☒ D. Data with missing values

Question ID : 441009619536
Option 1 ID : 4410092432295
Option 2 ID : 4410092432296
Option 3 ID : 4410092432298
Option 4 ID : 4410092432297
Status : Answered
Chosen Option : C

Q.8 What is the value for 'degrees of freedom' if the number of unknowns is less than the number of independent equations?

- Ans ☒ A. Zero
- ☒ B. Imaginary number
- ☒ C. Positive
- ☒ D. Negative

Question ID : 441009513047
Option 1 ID : 4410092007186
Option 2 ID : 4410092007187
Option 3 ID : 4410092007185
Option 4 ID : 4410092007184
Status : Answered
Chosen Option : D

Q.9 In a fuel cell, which of the following combinations ensures the highest efficiency and least environmental impact for stationary power generation?

- Ans ☒ A. Solid oxide fuel cell using hydrogen fuel
- ☒ B. Proton exchange membrane fuel cell using gasoline
- ☒ C. Phosphoric acid fuel cell using coal gas
- ☒ D. Methanol fuel cell with alkaline electrolyte

Question ID : 441009619440
Option 1 ID : 4410092431903
Option 2 ID : 4410092431906
Option 3 ID : 4410092431905
Option 4 ID : 4410092431904
Status : Answered
Chosen Option : A

Q.10 In semi-supervised learning, which of the following best describes the type of data used during training?

- Ans ☒ A. Neither labelled nor unlabelled data
- ☒ B. Only labelled data
- ☒ C. Both labelled and unlabelled data
- ☒ D. Only unlabelled data

Question ID : 441009619542
Option 1 ID : 4410092432322
Option 2 ID : 4410092432319
Option 3 ID : 4410092432321
Option 4 ID : 4410092432320
Status : Answered
Chosen Option : C

Q.11 In the context of momentum transfer in a fluid, what does the non-Newtonian behaviour of a fluid imply about its viscosity?

- Ans ☒ A. Viscosity is directly proportional to shear rate.
☒ B. Viscosity depends on the applied shear rate.
☒ C. Viscosity is constant regardless of shear rate.
☒ D. Viscosity is always zero.

Question ID : 441009641170

Option 1 ID : 4410092518816

Option 2 ID : 4410092518817

Option 3 ID : 4410092518815

Option 4 ID : 4410092518818

Status : Answered

Chosen Option : A

Q.12 The total energy emitted by a black body by radiation is 1200 W/m^2 . Calculate the absolute temperature of the body using the Stefan-Boltzmann constant as $5.67 \times 10^{-8} \text{ (W/m}^2 \text{ K}^4)$.

- Ans ☒ A. 334.26 K
☒ B. 472.81 K
☒ C. 381.42 K
☒ D. 513.74 K

Question ID : 441009513871

Option 1 ID : 4410092010814

Option 2 ID : 4410092010813

Option 3 ID : 4410092010812

Option 4 ID : 4410092010815

Status : Not Answered

Chosen Option : --

Q.13 Which profitability measure is defined as the annual percentage of net profit relative to the total capital investment, without considering the time value of money?

- Ans ☒ A. Net present value
☒ B. Discounted cash flow rate
☒ C. Simple rate of return
☒ D. Payback period

Question ID : 441009643372

Option 1 ID : 4410092527669

Option 2 ID : 4410092527666

Option 3 ID : 4410092527668

Option 4 ID : 4410092527667

Status : Answered

Chosen Option : A

Q.14 In controller tuning using the Ziegler-Nichols open-loop (process reaction curve) method, the ultimate period (Pu) and ultimate gain (Ku) are typically determined from:

- Ans ☒ A. the system's response to a sinusoidal input at varying frequencies
☒ B. a step change in the controller output to the final control element
☒ C. analysing the step response of the process itself (without control)
☒ D. applying varying proportional gains until continuous oscillations occur

Question ID : 441009620569

Option 1 ID : 4410092436409

Option 2 ID : 4410092436408

Option 3 ID : 4410092436410

Option 4 ID : 4410092436411

Status : Answered

Chosen Option : B

Q.15 Read the given statements carefully and select the correct option.

Statement A: For a pure substance at its triple point, the number of degrees of freedom (F) predicted by the Gibbs phase rule is zero.

Statement B: The Gibbs phase rule ($F = C - P + 2$) is applicable only to systems where all components are non-reactive and only one phase is present.

- Ans ☒ A. Both Statement A and Statement B are false.
☒ B. Statement A is false, but Statement B is true.
☒ C. Statement A is true, but Statement B is false.
☒ D. Both Statement A and Statement B are true.

Question ID : 441009642786

Option 1 ID : 4410092525284

Option 2 ID : 4410092525286

Option 3 ID : 4410092525285

Option 4 ID : 4410092525283

Status : Answered

Chosen Option : C

Q.16 High-density polyethylene (HDPE) is commonly produced using which type of polymerisation catalyst?

- Ans ☒ A. Anionic initiators
☒ B. Ziegler-Natta catalysts
☒ C. Cationic initiators
☒ D. Free radical initiators

Question ID : 441009644779

Option 1 ID : 4410092533221

Option 2 ID : 4410092533220

Option 3 ID : 4410092533222

Option 4 ID : 4410092533219

Status : Answered

Chosen Option : B

Q.17 Which of the following sectors is typically considered the hardest to decarbonise in the net zero roadmap?

- Ans ☒ A. Power generation
☒ B. Cement and steel industry
☒ C. Residential heating
☒ D. Agriculture

Question ID : 441009619418

Option 1 ID : 4410092431815

Option 2 ID : 4410092431816

Option 3 ID : 4410092431817

Option 4 ID : 4410092431818

Status : Answered

Chosen Option : D

Q.18 In a counterflow double-pipe heat exchanger, water and oil enter and exchange heat between them. Water is heated up and the oil temperature is reduced. The quantity of heat energy absorbed by water is 222.5 kW, overall heat transfer coefficient is 415 W/m² °C and LMTD is 41.57°C. Calculate the heat transfer area required.

- Ans ☒ A. 12.89 m²
☒ B. 10.54 m²
☒ C. 5.77 m²
☒ D. 27.63 m²

Question ID : 441009516148

Option 1 ID : 4410092019999

Option 2 ID : 4410092019998

Option 3 ID : 4410092020000

Option 4 ID : 4410092020001

Status : Not Answered

Chosen Option : --

Q.19 What is the monomer unit used to produce polyethylene?

- Ans ☒ A. Vinyl chloride
☒ B. Propylene
☒ C. Butadiene
☒ D. Ethylene

Question ID : 441009626318

Option 1 ID : 4410092459427

Option 2 ID : 4410092459424

Option 3 ID : 4410092459425

Option 4 ID : 4410092459426

Status : Answered

Chosen Option : D

Q.20 What is the typical rotating speed of Industrial paddle agitators?

- Ans ☒ A. 200 and 250 r/min
- ☒ B. 20 and 150 r/min
- ☒ C. 10 and 20 r/min
- ☒ D. 1 and 10 r/min

Question ID : 441009513494

Option 1 ID : 4410092009022

Option 2 ID : 4410092009023

Option 3 ID : 4410092009020

Option 4 ID : 4410092009021

Status : Not Answered

Chosen Option : --

Q.21 In a heat transfer process, what characteristic defines steady-state heat transfer?

- Ans ☒ A. Internal energy accumulates within the system.
- ☒ B. Temperatures at points remain constant.
- ☒ C. Heat flux changes dynamically across boundaries.
- ☒ D. System temperature continuously rises over time.

Question ID : 441009515884

Option 1 ID : 4410092018908

Option 2 ID : 4410092018909

Option 3 ID : 4410092018907

Option 4 ID : 4410092018906

Status : Answered

Chosen Option : B

Q.22 What is the meaning of 'Predictive Maintenance' in quality control?

- Ans ☒ A. Pilot testing and validation
- ☒ B. Predict equipment failure utilising past and present data
- ☒ C. Optimising the natural resources
- ☒ D. Improve the precision and effectiveness of quality control

Question ID : 441009562989

Option 1 ID : 4410092207077

Option 2 ID : 4410092207075

Option 3 ID : 4410092207076

Option 4 ID : 4410092207074

Status : Answered

Chosen Option : B

Q.23 Transport phenomena involves the study of transfer of which fundamental quantities?

- Ans ☒ A. Only heat and mass
- ☒ B. Momentum, energy, and mass
- ☒ C. Only momentum and energy
- ☒ D. Only chemical species

Question ID : 441009644886

Option 1 ID : 4410092533627

Option 2 ID : 4410092533629

Option 3 ID : 4410092533628

Option 4 ID : 4410092533630

Status : Answered

Chosen Option : B

Q.24 In a process stream with components A, B and C, if component C is a tie component, which of the following statements is true?

- Ans ☒ A. The flow rate of C is measured independently.
☒ B. Component C undergoes a chemical reaction.
☒ C. The flow rate of C is calculated using the flow rates of A and B.
☒ D. The flow rate of C is ignored in calculations.

Question ID : 441009619558

Option 1 ID : 4410092432387

Option 2 ID : 4410092432390

Option 3 ID : 4410092432388

Option 4 ID : 4410092432389

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.25 A chemical reactor has 8 variables and 6 independent equations including reaction and material balances. What is the degree of freedom?

- Ans ☒ A. 3
☒ B. 1
☒ C. 2
☒ D. 0

Question ID : 441009619574

Option 1 ID : 4410092432450

Option 2 ID : 4410092432448

Option 3 ID : 4410092432449

Option 4 ID : 4410092432447

Status : Answered

Chosen Option : C

Q.26 Polyvinyl chloride (PVC) is commonly used for pipes. Its synthesis involves the polymerisation of a monomer that contains which halogen element?

- Ans ☒ A. Chlorine
☒ B. Bromine
☒ C. Iodine
☒ D. Fluorine

Question ID : 441009628988

Option 1 ID : 4410092470105

Option 2 ID : 4410092470106

Option 3 ID : 4410092470107

Option 4 ID : 4410092470104

Status : Answered

Chosen Option : A

Q.27 If the activity coefficient (γ_i) for component i in a liquid mixture is equal to unity, this indicates that component i:

- Ans ☒ A. is completely immiscible with all other components that are present within the solution phase mixture
- ☒ B. possesses a fugacity in the mixture equal to its pure component fugacity at the system's T and P
- ☒ C. exists exclusively in its defined standard state, irrespective of the mixture composition or pressure
- ☒ D. behaves as if it were part of an ideal solution under the given conditions of composition and temperature

Question ID : 441009486036

Option 1 ID : 4410091899510

Option 2 ID : 4410091899511

Option 3 ID : 4410091899512

Option 4 ID : 4410091899509

Status : Answered

Chosen Option : D

Q.28 Fluid Catalytic Cracking (FCC) is a key process in petroleum refining primarily used to:

- Ans ☒ A. convert heavy fractions to lighter products
- ☒ B. increase naphtha octane using hydrogen
- ☒ C. desalt crude oil
- ☒ D. remove sulphur compounds from gasoline

Question ID : 441009644755

Option 1 ID : 4410092533128

Option 2 ID : 4410092533130

Option 3 ID : 4410092533127

Option 4 ID : 4410092533129

Status : Answered

Chosen Option : A

Q.29 Which of the following statements best describes the fundamental operating principle of a positive displacement pump, such as a gear pump or a piston pump?

- Ans ☒ A. It utilizes centrifugal force generated by a rotating impeller to accelerate the fluid radially outward to build pressure.
- ☒ B. It imparts velocity energy to the fluid, which is subsequently converted into pressure energy within the volute casing.
- ☒ C. It operates by trapping a fixed volume of fluid and then forcing that trapped volume into the discharge line.
- ☒ D. It relies on the Venturi effect to create a low-pressure zone, thereby drawing fluid into the pump from the suction side.

Question ID : 441009515779

Option 1 ID : 4410092018432

Option 2 ID : 4410092018430

Option 3 ID : 4410092018431

Option 4 ID : 4410092018433

Status : Answered

Chosen Option : B

Q.30 According to the National Fire Protection Association (NFPA) classification, a fire involving flammable liquids, greases, or gases are categorised as which class of fire?

- Ans ☒ A. Class D
☒ B. Class A
☒ C. Class C
☒ D. Class B

Question ID : 441009644856

Option 1 ID : 4410092533510

Option 2 ID : 4410092533507

Option 3 ID : 4410092533509

Option 4 ID : 4410092533508

Status : Answered

Chosen Option : D

Q.31 In the optimisation of a heat exchanger, increasing the hot fluid's flow velocity generally leads to which of the following?

- Ans ☒ A. A decrease in both heat transfer coefficient and pressure drop.
☒ B. An increase in the individual heat transfer coefficient and higher pressure drop.
☒ C. A decrease in the overall heat transfer coefficient and higher pressure drop.
☒ D. No significant change in heat transfer but a large decrease in pressure drop.

Question ID : 441009626222

Option 1 ID : 4410092459043

Option 2 ID : 4410092459041

Option 3 ID : 4410092459040

Option 4 ID : 4410092459042

Status : Answered

Chosen Option : B

Q.32 Aspen Plus is a widely used process simulation software. What is its primary application?

- Ans ☒ A. Computer-aided design (CAD) for mechanical parts
☒ B. Structural analysis of buildings
☒ C. Chemical process design and optimisation
☒ D. Financial accounting for chemical companies

Question ID : 441009642013

Option 1 ID : 4410092522195

Option 2 ID : 4410092522197

Option 3 ID : 4410092522196

Option 4 ID : 4410092522198

Status : Answered

Chosen Option : C

Q.33 Which type of distillation column and internal construction is most appropriate for handling high-volume, continuous processing operations?

- Ans ☒ A. Batch column with sieve plates
- ☒ B. Batch column with random packings
- ☒ C. Continuous column without any internal contacting devices
- ☒ D. Continuous column with bubble cap trays or valve trays

Question ID : 441009460840

Option 1 ID : 4410091802056

Option 2 ID : 4410091802059

Option 3 ID : 4410091802058

Option 4 ID : 4410091802057

Status : Answered

Chosen Option : C

Q.34 In a gas absorption tower of moderate size, how to minimise the channelling effect?

- Ans ☒ A. By maintaining large flow rates for liquid
- ☒ B. By covering the packing material surface by a non-stagnant film of liquid
- ☒ C. By filling the towers with stacked packing
- ☒ D. By having the diameter of the tower at least 8 times the packing diameter

Question ID : 441009516213

Option 1 ID : 4410092020288

Option 2 ID : 4410092020289

Option 3 ID : 4410092020286

Option 4 ID : 4410092020287

Status : Answered

Chosen Option : A

Q.35 Why is finding a minimum of the cost function essential in training ML models?

- Ans ☒ A. It allows the model to generalise better to unseen data.
- ☒ B. It maximises the number of features used.
- ☒ C. It helps in reducing the model accuracy.
- ☒ D. It increases the overfitting risk.

Question ID : 441009619510

Option 1 ID : 4410092432195

Option 2 ID : 4410092432198

Option 3 ID : 4410092432197

Option 4 ID : 4410092432196

Status : Answered

Chosen Option : C

Q.36 Which of the following is a major economic consideration in determining the optimum number of stages in a distillation column?

- Ans ☒ A. Trade-off between capital and operating costs
- ☒ B. Reboiler type
- ☒ C. Reflux drum size
- ☒ D. Tray spacing

Question ID : 441009589284

Option 1 ID : 4410092312256

Option 2 ID : 4410092312257

Option 3 ID : 4410092312254

Option 4 ID : 4410092312255

Status : Answered

Chosen Option : D

Q.37 In petroleum industry cracking reactions, which type of bond is broken?

- Ans ☒ A. Carbon – Nitrogen
☒ B. Hydrogen – Hydrogen
☒ C. Carbon – Carbon
☒ D. Carbon – Hydrogen

Question ID : 441009562570

Option 1 ID : 4410092205304

Option 2 ID : 4410092205303

Option 3 ID : 4410092205305

Option 4 ID : 4410092205302

Status : Answered

Chosen Option : C

Q.38 Which type of evaporator is characterised by a long tube bundle where liquid flows upwards, and the vapour formed causes a piston-like action, lifting the liquid and vapour mixture?

- Ans ☒ A. Short tube evaporator
☒ B. Long-tube vertical evaporator
☒ C. Plate evaporator
☒ D. Forced circulation evaporator

Question ID : 441009642955

Option 1 ID : 4410092525959

Option 2 ID : 4410092525961

Option 3 ID : 4410092525962

Option 4 ID : 4410092525960

Status : Answered

Chosen Option : B

Q.39 When simulating a highly recycled or reactive chemical process in Aspen Plus, which factor is most often the primary challenge to achieving a converged solution?

- Ans ☒ A. Defining basic components
☒ B. Good initial estimates
☒ C. Ensuring proper unit operation sequencing
☒ D. Visualising the PFD

Question ID : 441009644929

Option 1 ID : 4410092533799

Option 2 ID : 4410092533801

Option 3 ID : 4410092533800

Option 4 ID : 4410092533802

Status : Answered

Chosen Option : D

Q.40 According to American system of fire classifications, which category of fire is involved in burning metals?

- Ans ☒ A. Class A
☒ B. Class C
☒ C. Class B
☒ D. Class D

Question ID : 441009562682
Option 1 ID : 4410092205763
Option 2 ID : 4410092205765
Option 3 ID : 4410092205764
Option 4 ID : 4410092205762
Status : Answered
Chosen Option : B

Q.41 Which of the following is an advantage of adding integral action to a proportional controller (forming a PI controller)?

- Ans ☒ A. Reduces overshoot significantly.
☒ B. Increases system stability margins.
☒ C. Decreases system response time.
☒ D. Eliminates steady-state offset.

Question ID : 441009620537
Option 1 ID : 4410092436282
Option 2 ID : 4410092436283
Option 3 ID : 4410092436280
Option 4 ID : 4410092436281
Status : Answered
Chosen Option : D

Q.42 Compared to film theory, which of the following is a key characteristic of penetration theory in mass transfer?

- Ans ☒ A. It assumes a fixed, stagnant film thickness.
☒ B. Its mass-transfer coefficient is entirely independent of contact time.
☒ C. It involves steady-state molecular diffusion only.
☒ D. It addresses unsteady diffusion into fresh fluid elements.

Question ID : 441009516232
Option 1 ID : 4410092020347
Option 2 ID : 4410092020350
Option 3 ID : 4410092020348
Option 4 ID : 4410092020349
Status : Answered
Chosen Option : D

Q.43 When analysing the dynamic response of a transducer, what does the 'measurement lag' signify?

- Ans ☒ A. The maximum measurable range of the transducer.
- ☒ B. The time for output to reach its final steady value instantly.
- ☒ C. The random noise in the output signal.
- ☒ D. The delay between input change and output response.

Question ID : 441009643312

Option 1 ID : 4410092527417

Option 2 ID : 4410092527414

Option 3 ID : 4410092527416

Option 4 ID : 4410092527415

Status : Answered

Chosen Option : B

Q.44 A fluid has a flow curve where shear stress (τ) is plotted against shear rate ($\dot{\gamma}$), and the curve starts at a non-zero value of shear stress at zero shear rate and then increases linearly. What behaviour does this indicate?

- Ans ☒ A. Pseudoplastic behaviour
- ☒ B. Dilatant behaviour
- ☒ C. Bingham plastic behaviour
- ☒ D. Newtonian behaviour

Question ID : 441009459726

Option 1 ID : 4410091797573

Option 2 ID : 4410091797576

Option 3 ID : 4410091797575

Option 4 ID : 4410091797574

Status : Answered

Chosen Option : B

Q.45 The logarithmic mean concentration difference $(1-y)_{IM}$ is used in NTU calculations. Which of the following does it represent?

- Ans ☒ A. The effective driving force over the tower height
- ☒ B. The ratio of liquid to gas mass transfer coefficients
- ☒ C. The equilibrium solubility limit of the solute
- ☒ D. The arithmetic average of inlet and outlet mole fractions

Question ID : 441009460837

Option 1 ID : 4410091802037

Option 2 ID : 4410091802036

Option 3 ID : 4410091802038

Option 4 ID : 4410091802039

Status : Answered

Chosen Option : A

Q.46 In heterogeneous catalysis, where do the chemical reactions typically occur?

- Ans ☒ A. In the reactor walls.
- ☒ B. Only in the bulk liquid phase.
- ☒ C. On the catalyst surface.
- ☒ D. Only in the bulk gas phase.

Question ID : 441009620104

Option 1 ID : 4410092434540

Option 2 ID : 4410092434538

Option 3 ID : 4410092434539

Option 4 ID : 4410092434537

Status : Answered

Chosen Option : C

Q.47 In the analysis of heat transfer in multiphase systems involving boiling, which heat transfer regime is characterised by a significant reduction in the heat transfer coefficient due to a continuous vapour film insulating the heating surface?

- Ans ☒ A. Transition boiling
- ☒ B. Convective boiling
- ☒ C. Film boiling
- ☒ D. Nucleate boiling

Question ID : 441009641172

Option 1 ID : 4410092518824

Option 2 ID : 4410092518826

Option 3 ID : 4410092518825

Option 4 ID : 4410092518823

Status : Answered

Chosen Option : A

Q.48 What is the term for the boiling regime where bubbles form on a heated surface, detach, and rise, but do not yet form a continuous vapour film?

- Ans ☒ A. Nucleate boiling
- ☒ B. Film boiling
- ☒ C. Transition boiling
- ☒ D. Critical heat flux

Question ID : 441009642946

Option 1 ID : 4410092525925

Option 2 ID : 4410092525923

Option 3 ID : 4410092525924

Option 4 ID : 4410092525926

Status : Answered

Chosen Option : A

Q.49 Which type of tower is more suitable for operations involving high vacuum distillation due to lower gas pressure drop?

- Ans ☒ A. Plate towers only
- ☒ B. Packed towers only
- ☒ C. Neither packed towers nor plate towers
- ☒ D. Both Packed towers and Plate towers equally suitable

Question ID : 441009460826

Option 1 ID : 4410091801997

Option 2 ID : 4410091801996

Option 3 ID : 4410091801999

Option 4 ID : 4410091801998

Status : Answered

Chosen Option : D

Q.50 If the overall tray efficiency is 60%, and the number of theoretical stages required is 10, what is the approximate number of actual trays needed?

- Ans ☒ A. 10
- ☒ B. 17
- ☒ C. 6
- ☒ D. 21

Question ID : 441009460829

Option 1 ID : 4410091802009

Option 2 ID : 4410091802010

Option 3 ID : 4410091802008

Option 4 ID : 4410091802011

Status : Answered

Chosen Option : C

Q.51 When solving transport phenomena problems involving complex geometries or boundary conditions, which mathematical method is commonly used to approximate solutions by discretising the domain into a mesh of elements?

- Ans ☒ A. Dimensional analysis to find dimensionless groups
- ☒ B. Perturbation methods for small deviations
- ☒ C. Analytical solution by separation of variables
- ☒ D. Numerical solution using finite element method

Question ID : 441009641197

Option 1 ID : 4410092518925

Option 2 ID : 4410092518926

Option 3 ID : 4410092518923

Option 4 ID : 4410092518924

Status : Answered

Chosen Option : C

Q.52 Mass transfer coefficients are used to describe mass transfer in which regime?

- Ans ☒ A. Primarily convective flow.
- ☒ B. Primarily turbulent flow conditions.
- ☒ C. Only molecular diffusion in stagnant fluids.
- ☒ D. Only laminar flow without mixing.

Question ID : 441009643175

Option 1 ID : 4410092526868

Option 2 ID : 4410092526867

Option 3 ID : 4410092526866

Option 4 ID : 4410092526869

Status : Answered

Chosen Option : B

Q.53 Which characteristic distinguishes renewable energy sources from fossil fuels?

- Ans ☒ A. They are primary sources of greenhouse gas emissions.
- ☒ B. They have a finite supply that depletes with use.
- ☒ C. Their generation always requires chemical conversion.
- ☒ D. They are naturally replenished on a human timescale.

Question ID : 441009641370

Option 1 ID : 4410092519620

Option 2 ID : 4410092519619

Option 3 ID : 4410092519622

Option 4 ID : 4410092519621

Status : Answered

Chosen Option : B

Q.54 During dropwise condensation, the heat transfer coefficient is higher than filmwise condensation because:

- Ans ☒ A. the area available for condensation is reduced
- ☒ B. the vapour velocity is higher
- ☒ C. the condensate is removed faster, minimising thermal resistance
- ☒ D. the surface temperature is lower

Question ID : 441009460759

Option 1 ID : 4410091801740

Option 2 ID : 4410091801743

Option 3 ID : 4410091801741

Option 4 ID : 4410091801742

Status : Answered

Chosen Option : A

Q.55 Which of the following is considered a deflagration?

- Ans ☒ A. A detonation with a supersonic shock front
- ☒ B. A slow combustion wave travelling below the speed of sound
- ☒ C. A sudden burst of steam
- ☒ D. An electrical arc

Question ID : 441009611027

Option 1 ID : 4410092399232

Option 2 ID : 4410092399231

Option 3 ID : 4410092399230

Option 4 ID : 4410092399229

Status : Answered

Chosen Option : C

Q.56 In Carbon Capture and Storage (CCS), which technology is most commonly employed for post-combustion CO₂ capture from industrial flue gases?

- Ans ☒ A. Direct air capture
- ☒ B. Amine-based absorption
- ☒ C. Membrane separation
- ☒ D. Cryogenic separation

Question ID : 441009641428

Option 1 ID : 4410092519854

Option 2 ID : 4410092519853

Option 3 ID : 4410092519852

Option 4 ID : 4410092519851

Status : Answered

Chosen Option : A

Q.57 Read the given statements carefully and select the correct option.

Statement A: A significant risk of using the payback period method for investment decisions is that it typically disregards the time value of money.

Statement B: Projects with a longer payback period are always economically superior to those with a shorter payback period, irrespective of cash flows occurring beyond the payback period.

- Ans ☒ A. Statement A is false, but Statement B is true.
- ☒ B. Both Statement A and Statement B are false.
- ☒ C. Both Statement A and Statement B are true.
- ☒ D. Statement A is true, but Statement B is false.

Question ID : 441009643424

Option 1 ID : 4410092527881

Option 2 ID : 4410092527879

Option 3 ID : 4410092527878

Option 4 ID : 4410092527880

Status : Not Answered

Chosen Option : --

Q.58 Which flow meter works based on the pressure difference created by an obstruction in the flow?

- Ans ☒ A. Ultrasonic flow meter
- ☒ B. Orifice meter
- ☒ C. Rotameter
- ☒ D. Electromagnetic flow meter

Question ID : 441009459755

Option 1 ID : 4410091797678

Option 2 ID : 4410091797677

Option 3 ID : 4410091797679

Option 4 ID : 4410091797680

Status : Answered

Chosen Option : C

Q.59 Which of the following statements accurately describes the nature and behaviour of a thermal boundary layer in forced convection?

- Ans ☒ A. Its thickness is independent of fluid properties and flow conditions at the interface.
- ☒ B. It is a region where temperature varies from surface to bulk.
- ☒ C. Heat transfer is negligible due to the complete absence of any fluid motion in this region.
- ☒ D. It is a region where fluid velocity is uniform and heat transfer occurs solely by conduction.

Question ID : 441009515940

Option 1 ID : 4410092019171

Option 2 ID : 4410092019172

Option 3 ID : 4410092019173

Option 4 ID : 4410092019170

Status : Answered

Chosen Option : B

Q.60 Which of the following best describes the primary function of a transducer in a process control system?

- Ans ☒ A. Comparing a measured value to a set point.
- ☒ B. Storing process data for later analysis.
- ☒ C. Converts a process variable into a signal.
- ☒ D. Directly manipulating a final control element.

Question ID : 441009620137

Option 1 ID : 4410092434678

Option 2 ID : 4410092434679

Option 3 ID : 4410092434677

Option 4 ID : 4410092434676

Status : Answered

Chosen Option : C

Q.61 What is the meaning of residual Gibbs energy?

- Ans ☒ A. Difference between the actual and ideal values of compressibility factor
- ☒ B. Difference between the actual and ideal gas values of Gibbs energy at the same temperature and pressure
- ☒ C. Difference between the Helmholtz free energy and Gibbs energy at the same temperature and pressure
- ☒ D. Sum of the actual and ideal gas values of Gibbs energy at the same temperature and pressure

Question ID : 441009513109

Option 1 ID : 4410092007460

Option 2 ID : 4410092007461

Option 3 ID : 4410092007463

Option 4 ID : 4410092007462

Status : Answered

Chosen Option : B

Q.62 For an elementary liquid-phase reaction $A + B \rightarrow R + S$, to maximize the yield of R in a system where B is expensive and conversion of A is high, which ideal reactor configuration is generally preferred?

- Ans ☒ A. A single Continuous Stirred Tank Reactor (CSTR).
☒ B. A PFR followed by a CSTR.
☒ C. A single Plug Flow Reactor (PFR).
☒ D. Multiple CSTRs in series.

Question ID : 441009620070

Option 1 ID : 4410092434397

Option 2 ID : 4410092434400

Option 3 ID : 4410092434398

Option 4 ID : 4410092434399

Status : Answered

Chosen Option : C

Q.63 Which type of fire extinguisher is generally most effective for extinguishing Class B fires involving flammable liquids or gases?

- Ans ☒ A. Dry chemical extinguisher
☒ B. Water extinguisher
☒ C. Carbon dioxide (CO₂) extinguisher
☒ D. Foam extinguisher

Question ID : 441009637709

Option 1 ID : 4410092504980

Option 2 ID : 4410092504979

Option 3 ID : 4410092504981

Option 4 ID : 4410092504982

Status : Answered

Chosen Option : C

Q.64 A variable that is used to describe a family of solutions is a _____.

- Ans ☒ A. forcing function
☒ B. Eigen vector
☒ C. parameter
☒ D. response variable

Question ID : 441009562947

Option 1 ID : 4410092206900

Option 2 ID : 4410092206901

Option 3 ID : 4410092206899

Option 4 ID : 4410092206898

Status : Answered

Chosen Option : B

Q.65 What is the typical particle size measurement range of standard screens?

- Ans ☒ A. 10 nm and 100 μm
☒ B. 76 mm and 38 μm
☒ C. 500 μm and 80 mm
☒ D. 0.01 mm and 1 cm

Question ID : 441009513458

Option 1 ID : 4410092008881

Option 2 ID : 4410092008880

Option 3 ID : 4410092008883

Option 4 ID : 4410092008882

Status : Answered

Chosen Option : D

Q.66 What will happen to fluid temperature, when the pressure on a compressible fluid is increased adiabatically?

- Ans ☒ A. The temperature of the fluid also increases
- ☒ B. The temperature of the fluid remains constant
- ☒ C. The temperature of the fluid decreases exponentially
- ☒ D. The temperature of the fluid varies randomly and no specific pattern is observed

Question ID : 441009513861

Option 1 ID : 4410092010773

Option 2 ID : 4410092010774

Option 3 ID : 4410092010772

Option 4 ID : 4410092010775

Status : Answered

Chosen Option : A

Q.67 Which of the following phenomena is a characteristic of non-ideal flow in chemical reactors?

- Ans ☒ A. Presence of stagnant regions or bypassing.
- ☒ B. All fluid elements having identical residence times.
- ☒ C. Perfect mixing throughout the reactor volume.
- ☒ D. Uniform velocity profile across the reactor cross-section.

Question ID : 441009620078

Option 1 ID : 4410092434431

Option 2 ID : 4410092434432

Option 3 ID : 4410092434429

Option 4 ID : 4410092434430

Status : Answered

Chosen Option : A

Q.68 When interpreting kinetic data, if a reaction exhibits a straight line on a plot of $1/C_A$ versus time, what is the most likely order of the reaction?

- Ans ☒ A. Second-order
- ☒ B. Zero-order
- ☒ C. First-order
- ☒ D. Third-order

Question ID : 441009620016

Option 1 ID : 4410092434195

Option 2 ID : 4410092434193

Option 3 ID : 4410092434194

Option 4 ID : 4410092434196

Status : Answered

Chosen Option : B

Q.69 Which of the following cost categories is typically classified as a fixed cost in chemical plant economics?

- Ans ☒ A. Raw material cost
☒ B. Operator wages
☒ C. Depreciation
☒ D. Maintenance cost

Question ID : 441009583544

Option 1 ID : 4410092289302

Option 2 ID : 4410092289303

Option 3 ID : 4410092289305

Option 4 ID : 4410092289304

Status : Answered

Chosen Option : D

Q.70 For a continuous counter current leaching operation, in which direction are the stages numbered?

- Ans ☒ A. In the direction of liquid flow
☒ B. In the direction of flow of the solid
☒ C. Lateral direction
☒ D. Random directions

Question ID : 441009516224

Option 1 ID : 4410092020319

Option 2 ID : 4410092020318

Option 3 ID : 4410092020321

Option 4 ID : 4410092020320

Status : Answered

Chosen Option : C

Q.71 A straight line tangent is drawn to the process reaction curve at the point of inflection. Which parameter value is obtained when the tangent line intersects with the time axis?

- Ans ☒ A. Steady state gain
☒ B. Minimum offset
☒ C. Apparent transport lag
☒ D. Real valued time constant

Question ID : 441009526117

Option 1 ID : 4410092059884

Option 2 ID : 4410092059887

Option 3 ID : 4410092059886

Option 4 ID : 4410092059885

Status : Answered

Chosen Option : A

Q.72 Why are iterative calculations often required to solve process models, especially for complex systems?

- Ans ☒ A. To achieve faster convergence
☒ B. To reduce model complexity
☒ C. For numerical stability
☒ D. Due to non-linear equations

Question ID : 441009644922

Option 1 ID : 4410092533774

Option 2 ID : 4410092533771

Option 3 ID : 4410092533772

Option 4 ID : 4410092533773

Status : Answered

Chosen Option : D

Q.73 In which of the following conditions can the quality of control often be improved by the addition of feedforward control system?

- Ans ☒ A. If the error value is zero
☒ B. If there is a random fluctuation in set point value
☒ C. If a particular load disturbance occurs frequently in a control process
☒ D. If the damping term is dominating in the response

Question ID : 441009526165

Option 1 ID : 4410092060079

Option 2 ID : 4410092060078

Option 3 ID : 4410092060077

Option 4 ID : 4410092060076

Status : Not Answered

Chosen Option : --

Q.74 Consider a pipe flow where fluid velocity increases from 2 m/s to 4 m/s. Assuming no pump or turbine and negligible losses, what happens to the pressure, according to Bernoulli's equation?

- Ans ☒ A. Pressure increases
☒ B. Pressure remains constant
☒ C. Pressure decreases
☒ D. Pressure varies depending on pipe diameter only

Question ID : 441009459747

Option 1 ID : 4410091797649

Option 2 ID : 4410091797651

Option 3 ID : 4410091797650

Option 4 ID : 4410091797652

Status : Answered

Chosen Option : C

Q.75 Which of the following is the primary purpose of Residence Time Distribution (RTD) analysis?

- Ans ☒ A. To determine reactor temperature
- ☒ B. To measure reaction order
- ☒ C. To calculate catalyst effectiveness
- ☒ D. To study flow behaviour in reactors

Question ID : 441009515943

Option 1 ID : 4410092019191

Option 2 ID : 4410092019193

Option 3 ID : 4410092019190

Option 4 ID : 4410092019192

Status : Answered

Chosen Option : D

Q.76 A U-tube manometer contains mercury (density ρ_m) and is used to measure the pressure of a gas in a tank. The mercury level on the side open to the atmosphere is higher by a distance h than the level on the side connected to the tank. If atmospheric pressure is P_{atm} , the absolute pressure of the gas (P_{gas}) is best expressed by:

- Ans ☒ A. $P_{gas} = \rho_m gh$, considering only the manometric fluid's contribution to total gauge pressure.
- ☒ B. $P_{gas} = P_{atm} + \rho_m gh$, because the higher mercury column directly adds to atmospheric pressure.
- ☒ C. $P_{gas} = P_{atm} - \rho_m gh$, since the atmospheric pressure pushes the mercury column down on one side.
- ☒ D. $P_{gas} = \rho_m gh - P_{atm}$, as the manometric fluid pressure must overcome the atmospheric pressure.

Question ID : 441009486262

Option 1 ID : 4410091900432

Option 2 ID : 4410091900429

Option 3 ID : 4410091900430

Option 4 ID : 4410091900431

Status : Answered

Chosen Option : B

Q.77 According to the National Fire Protection Association (NFPA) classification, a fire involving ordinary combustible material like wood, paper, and cloth is classified as which class of fire?

- Ans ☒ A. Class A
- ☒ B. Class D
- ☒ C. Class C
- ☒ D. Class B

Question ID : 441009637696

Option 1 ID : 4410092504929

Option 2 ID : 4410092504930

Option 3 ID : 4410092504928

Option 4 ID : 4410092504927

Status : Answered

Chosen Option : A

Q.78 What is the autoignition temperature of a substance defined as?

- Ans ☒ A. The highest temperature where it will not ignite in air.
- ☒ B. Temperature at which a liquid completely vaporises.
- ☒ C. Lowest temp for spontaneous ignition without external source.
- ☒ D. The minimum temperature to sustain combustion.

Question ID : 441009640569

Option 1 ID : 4410092516432

Option 2 ID : 4410092516434

Option 3 ID : 4410092516433

Option 4 ID : 4410092516431

Status : Answered

Chosen Option : C

Q.79 In process flow modeling, the fundamental relationships describing equipment behaviour and material/energy transformations are typically expressed as:

- Ans ☒ A. detailed piping diagrams
- ☒ B. sets of algebraic equations
- ☒ C. empirical correlations
- ☒ D. graphical sketches only

Question ID : 441009644921

Option 1 ID : 4410092533770

Option 2 ID : 4410092533769

Option 3 ID : 4410092533768

Option 4 ID : 4410092533767

Status : Answered

Chosen Option : B

Q.80 What is the main goal of unsupervised learning?

- Ans ☒ A. To discover hidden patterns in unlabeled data
- ☒ B. To classify data into predefined categories
- ☒ C. To learn from labeled data with human supervision
- ☒ D. To predict a continuous output variable

Question ID : 441009642285

Option 1 ID : 4410092523285

Option 2 ID : 4410092523284

Option 3 ID : 4410092523286

Option 4 ID : 4410092523283

Status : Answered

Chosen Option : C

Q.81 The time constant (τ) of a first-order system indicates ____.

- Ans ☒ A. the system gain
- ☒ B. the time for the system to return to initial state
- ☒ C. the time for the output to reach 63.2% of its total change
- ☒ D. the time at which the output reaches 100% of the final value

Question ID : 441009576424

Option 1 ID : 4410092260821

Option 2 ID : 4410092260820

Option 3 ID : 4410092260819

Option 4 ID : 4410092260818

Status : Answered

Chosen Option : C

Q.82 When using the initial rate method for kinetic data interpretation, what is primarily determined from varying the initial concentration of a reactant while holding others constant?

- Ans ☒ A. The reaction order
☐ B. The rate constant
☐ C. The activation energy
☐ D. The frequency factor

Question ID : 441009620061
Option 1 ID : 4410092434361
Option 2 ID : 4410092434364
Option 3 ID : 4410092434362
Option 4 ID : 4410092434363
Status : Answered
Chosen Option : A

Q.83 In industrial production, what is the typical operating pressure range for gas-phase polymerisation of polypropylene?

- Ans ☐ A. 1-2 atm
☒ B. 10-30 atm
☐ C. 150-200 atm
☐ D. 60-100 atm

Question ID : 441009605595
Option 1 ID : 4410092377497
Option 2 ID : 4410092377498
Option 3 ID : 4410092377500
Option 4 ID : 4410092377499
Status : Answered
Chosen Option : B

Q.84 Which numerical method is commonly used in process simulators for solving nonlinear algebraic equations?

- Ans ☐ A. Runge-Kutta method
☒ B. Newton-Raphson method
☐ C. Euler's method
☐ D. Simpson's rule

Question ID : 441009620502
Option 1 ID : 4410092436142
Option 2 ID : 4410092436141
Option 3 ID : 4410092436143
Option 4 ID : 4410092436140
Status : Answered
Chosen Option : D

Q.85 Evaporation is a process where liquid changes into vapour, and it requires:

- Ans ☒ A. release of latent heat from the system
- ☒ B. complete vacuum conditions for efficient operation
- ☒ C. latent heat absorption
- ☒ D. a significant increase in local pressure

Question ID : 441009516116

Option 1 ID : 4410092019870

Option 2 ID : 4410092019873

Option 3 ID : 4410092019871

Option 4 ID : 4410092019872

Status : Answered

Chosen Option : A



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